Date: 02/02/2019

Department: Software Engineering

Course: Event-Driven Programming

Group Project — C# Desktop → API/Data → Web

Group size: 5–6 students
Duration: 2 months
Phases & Marks:

- Desktop Application (WinForms): **5 marks**
- API + Data Layer: **5 marks**
- Web Application: **5 marks**
- Group participation (individual contribution): **4 marks**
  Total: 15 marks

Students work in phases. Each phase is graded even if later improvements are added.

1. Project Process

Phase 1 Desktop Application (Weeks 1–3)

- Event-driven UI (buttons, forms, validation, navigation)
- Local memory storage (no database required yet)
- Basic business logic
- Clean UI and structure

Grading after Phase 1

Phase 2 API + Data Layer (Weeks 4–6)

- Create REST API (ASP.NET Web API)
- Connect Desktop app to API
- Add database (SQL Server/PostgreSQL)
- Implement CRUD operations

Grading after Phase 2

Phase 3 Web Application (Weeks 7–8)

- Web UI (ASP.NET MVC / Razor / minimal frontend)
- Connect to same API
- Authentication (basic)
- Reports or dashboards
- You can use AI for the web app.

Final grading

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2. Student Deliverables

Each group must submit:

- Source code (Desktop + API + Web)
- Database schema
- Simple documentation (5–10 pages)
- Group contribution report

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3. Project Selection List (20 Options)

Students choose **ONE**.

1. Student Course Registration System
2. Library Management System
3. Hotel Reservation System
4. Clinic/Patient Appointment System
5. Inventory Management System
6. Pharmacy Management System
7. Restaurant Order Management
8. Cinema Ticket Booking System
9. Gym Membership System
10. School Fee Management System
11. Vehicle Rental System
12. Event Registration System
13. Employee Leave Management
14. Online Quiz Management System
15. Supermarket POS System
16. Bookstore Management System
17. Bus Ticket Booking System
18. Job Application Tracking System
19. Meeting Room Reservation System
20. University Club Management System

All are sized to be completed in **2 months**, the web part should be big and complex as you develop it using AI.

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4. Functional Requirements (Instructor Reference)

Students must propose their own features first.
Instructor will approve based on the list below.

Core Desktop Requirements

- Multi-form navigation
- Event-driven controls
- Input validation
- Data entry forms
- List display (ListView)
- Search/filter
- Edit & delete actions
- Menu/toolbar
- Status messages
- Simple reporting view

API + Data Layer Requirements

- REST API (CRUD)
- Database tables
- Entity relationships
- Repository/service layer
- Error handling
- Logging (basic)
- DTO usage
- Connection from Desktop to API

Web Application Requirements

- Login page
- Dashboard
- CRUD pages
- API integration

- Search/filter
- Responsive layout
- Simple report page

Advanced Features (Optional for higher score)

- Role-based login
- Notifications
- Charts
- Export to PDF
- Settings panel
- Audit logs

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5. Group Participation (4 marks)

Given individually based on:

- Contribution
- Git commits
- Task completion
- Team cooperation

Instructor may ask each student questions.

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6. Submission Timeline

Date	Task
6/02/2019	Project selection with list of functionalities
16/02/2019	Desktop demo
29/03/2019	API + DB demo
18/04/2019	Final web demo

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7. Rules

- One project per group
- Must follow phases
- UI must be event-driven
- Copying between groups not allowed

- Late submission reduces marks
- Using may reduces marks